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To cite this article: Gabriel Hadjerci *et al* 2026 *J. Phys.: Conf. Ser.* **3230** 012026

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Steady solutions for the analysis of scaling regimes in rapidly rotating Rayleigh–Bénard convection

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Abstract. Heat transport and velocity statistics in rapidly rotating Rayleigh–Bénard convection with stress-free boundaries are examined. The theoretical ‘ultimate’ scaling regime of rapidly rotating convection predicts that the Nusselt and Reynolds numbers become independent of the small molecular diffusivities for very high Rayleigh number. However, such diffusivity-free behavior has been observed only for the Nusselt number, while the Reynolds number deviates from this prediction. In this study, we employ the numerical Newton–Krylov method to track a continuous branch of steady solutions to an asymptotically reduced system of equations for rapidly rotating convection. We show that, along this branch, the Nusselt and Reynolds numbers are related in such a way that both cannot simultaneously exhibit diffusivity-free scaling. Based on this relationship, we derive a new scaling prediction for the Reynolds number, assuming a diffusivity-free scaling law for the Nusselt number. Direct numerical simulations of both the asymptotically reduced system and the full Boussinesq equations show excellent agreement with this prediction, confirming its relevance even to the fully chaotic regime.

1 Introduction

Thermal convection under rapid global rotation is a key process in geophysical and astrophysical systems. In particular, it can drive strong turbulent flows in stellar interiors, planetary oceans, and atmospheres. These flows, in turn, may sustain magnetic fields or influence planetary climates [1–4]. Quantifying heat and momentum transport in rotating turbulent convection is therefore of fundamental interest. However, the Reynolds numbers in such systems can reach extremely large values—far exceeding those attainable in laboratory experiments or numerical simulations. To address this limitation, one usually relies on scaling laws derived from dimensional analysis or turbulence modeling. Within the framework of turbulence theory, the assumption of fully developed turbulence has led to the concept of an ‘ultimate’ scaling regime of convection. In this regime, consistent with the energy transfer laws of turbulence, the large-scale properties of the flow are expected to become independent of the small molecular diffusivities, such as viscosity and thermal diffusivity [2, 5, 6]. In other words, diffusivity-free scaling laws are expected for the convective heat flux and momentum transport, characterized respectively by the Nusselt number Nu and the Reynolds number Re .

The ultimate regime of rapidly rotating convection, also referred to as the geostrophic turbulence scaling regime [7, 8], has been observed and characterized in a few recent studies, including direct numerical simulations (DNS) and laboratory experiments [8–11]. Among the configurations of rotating convection, rotating Rayleigh–Bénard convection (RRBC) is by far the most extensively studied case [12–14]. In this configuration, the fluid is confined between two horizontal plates, with the lower plate maintained at a



higher temperature than the upper one, thereby driving convection between them. The precise conditions under which the ultimate regime can be observed remain uncertain. In particular, a recent study by Kannan et al. [15] demonstrated that the velocity boundary conditions at the plates play a crucial role in determining the emergence of the ultimate regime in RRBC. With no-slip boundaries, diffusivity-free scaling behavior is observed for both the Nusselt and Reynolds numbers. In contrast, under stress-free boundary conditions, diffusivity-free scaling applies only to the Nusselt number, while the Reynolds number deviates from this prediction. This result is surprising, as one would normally expect the kinematic boundary layers associated with no-slip boundaries to introduce viscous effects into the scaling, rather than suppress them.

The observations in stress-free RRBC deviate from the predictions of the ultimate scaling regime of rapidly rotating convection. Kannan et al. [15] attempted to explain this discrepancy using phenomenological arguments. Indeed, when studying turbulent flows, phenomenological reasoning is the standard approach, as the inherent complexity of turbulence makes it difficult to obtain rigorous results directly from the governing equations of motion. In this context, invariant solutions become particularly valuable: although they do not represent the fully chaotic state, they can capture key features of it or provide rigorous bounds on its scaling properties [16–20]. Moreover, unlike turbulent solutions, invariant solutions allow for the derivation of exact and rigorous relations [21]. In this study, we first track a continuous branch of steady solutions to an asymptotically reduced system of equations describing rotationally constrained Rayleigh-Bénard convection with stress-free boundary conditions. We then demonstrate that, along this branch, the Nusselt and Reynolds numbers are related in such a way that both cannot simultaneously exhibit diffusivity-free scaling. Finally, assuming that this incompatibility property extends to the chaotic solutions, we derive a scaling prediction for the Reynolds number that is in good agreement with both DNS of the reduced system of equations and DNS of the full Navier-Stokes equations.

2 The asymptotic reduced model of rapidly rotating Rayleigh-Bénard convection

2.1 Rotating Rayleigh-Bénard convection

A fluid layer of height H is confined between two horizontal bounding surfaces. The system is subject to vertical gravity $-g\mathbf{e}_z$ and undergoes global rotation with constant angular velocity $\Omega\mathbf{e}_z$ according to an inertial frame. The lower surface is maintained at a temperature higher than that of the upper surface by ΔT , thereby driving convection once ΔT is sufficiently large. We adopt the Boussinesq approximation [22], under which all fluid properties are taken as constant and uniform, except for the density, whose linear dependence on temperature is retained in the buoyancy force. Our analysis is further restricted to parameter regimes where centrifugal acceleration is negligible [23, 24]. The fluid is characterized by a kinematic viscosity ν , a thermal diffusivity κ , and a thermal expansion coefficient α . Dimensionless governing equations are obtained by measuring length in units of H , time in units of H^2/ν , and temperature in units of ΔT . This nondimensionalization introduces three control parameters: the Rayleigh number, the Prandtl number, and the Ekman number:

$$Ra = \frac{\alpha g \Delta T H^3}{\nu \kappa}; \quad Pr = \frac{\nu}{\kappa}; \quad E = \frac{\nu}{2\Omega H^2}. \quad (1)$$

Denoting $\mathbf{u} = u\mathbf{e}_x + v\mathbf{e}_y + w\mathbf{e}_z$ the dimensionless velocity field, p the dimensionless pressure, and Θ the dimensionless temperature, equations of motions are:

$$\nabla \cdot \mathbf{u} = 0, \quad (2a)$$

$$\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} + E^{-1} \mathbf{e}_z \times \mathbf{u} = -\nabla p + Ra Pr^{-1} [\Theta - (1 - z)] \mathbf{e}_z + \nabla^2 \mathbf{u}, \quad (2b)$$

$$\partial_t \Theta + \mathbf{u} \cdot \nabla \Theta = Pr^{-1} \nabla^2 \Theta. \quad (2c)$$

All the fields are supposed periodic in the horizontal directions. Temperature is fixed at the top and bottom, such that:

$$\Theta(z=0) = 1 \quad \text{and} \quad \Theta(z=1) = 0. \quad (3)$$

Velocity boundary conditions appeared to be crucial for RRBC scaling regimes. In this article, stress-free boundaries with no-penetrability condition are discussed, i.e. at $z=0$ and $z=1$:

$$\partial_z u = \partial_z v = 0 \quad \text{and} \quad w = 0. \quad (4)$$

In order to eliminate the pressure field, the mass conservation is leveraged to introduce the poloidal-toroidal decomposition of the velocity field:

$$\mathbf{u} = \nabla \times \nabla \times (\phi \mathbf{e}_z) + \nabla \times (\psi \mathbf{e}_z) + \bar{u}(z)\mathbf{e}_x + \bar{v}(z)\mathbf{e}_y, \quad (5)$$

where ψ is the toroidal stream function, ϕ the poloidal potential, and, $\bar{u}(z)$ and $\bar{v}(z)$ are mean flows that depend only on depth. The pressure gradient is then eliminated by taking the curl of the momentum equation.

2.2 The asymptotic reduced model

Based on physical arguments, Julien et al. [25] derived an asymptotic expansion of the equations of RRBC in the limit of asymptotically rapid rotation, or equivalently, asymptotically small Ekman number. A key ingredient of their approach is the double-scale separation in both time and space. Let ε be an arbitrary small parameter. The model introduces small-scale spatial coordinates (x, y, z) measured in units of εH , together with a large-scale vertical coordinate Z measured in units of H . A similar separation holds for time: the model features a fast time variable t measured in units of $\varepsilon^2 H^2 / \nu$ and a slow time variable T measured in units of H^2 / ν . With these new units in place, we choose to nondimensionalize the velocities with $\nu / (\varepsilon H)$. We then decompose the fields into mean and fluctuating parts by defining an average over the fluctuations:

$$\bar{\chi}(Z, T) \stackrel{\text{def}}{=} \lim_{\tau, V \rightarrow \infty} \frac{1}{\tau V} \int_{\tau, V} \chi(x, y, z, Z, t, T) \, dx \, dy \, dz \, dt, \quad (6)$$

for any dependent quantity χ . The fluctuating component χ' is therefore defined by:

$$\chi(x, y, z, Z, t, T) = \bar{\chi}(Z, T) + \chi'(x, y, z, Z, t, T). \quad (7)$$

Julien et al. [25] provide arguments for performing the expansion with $\varepsilon = E^{1/3}$ as the small parameter. In this framework, they showed that the asymptotic expansions of the fluctuating velocity components (u', v', w') and of the mean and fluctuating temperatures $\bar{\Theta}$ and Θ' can be written in the form:

$$u' = \partial_y \psi + \mathcal{O}(\varepsilon), \quad (8a)$$

$$v' = -\partial_x \psi + \mathcal{O}(\varepsilon), \quad (8b)$$

$$w' = -\nabla_{\perp}^2 \phi + \mathcal{O}(\varepsilon), \quad (8c)$$

$$\bar{\Theta} = \bar{\theta} + \mathcal{O}(\varepsilon), \quad (8d)$$

$$\Theta' = 0 + \varepsilon \theta' + \mathcal{O}(\varepsilon^2), \quad (8e)$$

where $\nabla_{\perp}$ denotes the horizontal gradient, and where $(\psi, \phi, \theta', \bar{\theta})$ are variables independent of the small vertical coordinate z . At higher orders, one additionally obtains:

$$\partial_y u' - \partial_x v' = -\zeta = \nabla_{\perp}^2 \psi + \mathcal{O}(\varepsilon), \quad (9a)$$

$$\partial_x u' + \partial_y v' = -\varepsilon \partial_Z w' = 0 + \varepsilon \nabla_{\perp}^2 (\partial_Z \phi) + \mathcal{O}(\varepsilon^2), \quad (9b)$$

where ζ is the vertical vorticity. Finally, the dynamics of RRBC in the fast rotation limit is described by the asymptotic reduced model (ARM) provided by Sprague et al. [26]:

$$\partial_t \nabla_{\perp}^2 \psi = J(\psi, \nabla_{\perp}^2 \psi) + \partial_Z \nabla_{\perp}^2 \phi + \nabla_{\perp}^4 \psi, \quad (10a)$$

$$\partial_t \nabla_{\perp}^2 \phi = J(\psi, \nabla_{\perp}^2 \phi) - \partial_Z \psi - \frac{\widetilde{Ra}}{Pr} \theta' + \nabla_{\perp}^4 \phi, \quad (10b)$$

$$\partial_t \theta' = J(\psi, \theta') + \nabla_{\perp}^2 \phi \partial_Z \bar{\theta} + Pr^{-1} \nabla_{\perp}^2 \theta', \quad (10c)$$

$$\partial_T \bar{\theta} = \partial_Z (\overline{\theta' \nabla_{\perp}^2 \phi}) + Pr^{-1} \partial_{ZZ} \bar{\theta}, \quad (10d)$$

where $\widetilde{Ra} = Ra E^{4/3}$ is the reduced Rayleigh number and where $J(f, g) = \partial_x f \partial_y g - \partial_y f \partial_x g$. Impermeability and fixed temperature boundaries can be written as:

$$\phi|_{Z=0} = \phi|_{Z=1} = 0, \quad (11a)$$

$$\theta'|_{Z=0} = \theta'|_{Z=1} = 0, \quad (11b)$$

$$\bar{\theta}|_{Z=0} = 1 \quad \text{and} \quad \bar{\theta}|_{Z=1} = 0. \quad (11c)$$

Those boundary conditions, when inserted into the set of equations (10), involve other boundary conditions:

$$\partial_Z \psi|_{Z=0} = \partial_Z \psi|_{Z=1} = 0, \quad (12a)$$

$$\partial_{ZZ} \phi|_{Z=0} = \partial_{ZZ} \phi|_{Z=1} = 0. \quad (12b)$$

In particular, stress-free boundary conditions (12a) appear to be an implication of impermeability in the ARM and not a given boundary condition.

2.3 The ultimate regime of rapidly rotating Rayleigh–Bénard convection

An interesting feature of the ARM is that it depends on only two control parameters, whereas RRBC involves three. Specifically, the dependence on the Rayleigh and Ekman numbers is combined into a single parameter, the reduced Rayleigh number $\widetilde{Ra} = RaE^{4/3}$. This parameter is often referred to as the supercriticality parameter, since in the asymptotic limit $E \rightarrow 0$, the critical Rayleigh number at the onset of convection, Ra_c , follows:

$$Ra_c \sim E^{-4/3}, \quad (13)$$

in other words $\widetilde{Ra} \sim Ra/Ra_c$. When $Ra > Ra_c$, the conductive state becomes unstable and the flow exhibits convective heat flux. After some transient, the system reaches a statistically stationary state. The focus of this study is the statistical properties of this state.

The Nusselt number is defined as the ratio between the total heat flux and the conductive heat flux. Thus, the instantaneous Nusselt number expressed in terms of the ARM quantities is:

$$Nu = 1 + Pr \langle w'\theta' \rangle_{V,t}, \quad (14)$$

where $\langle \cdot \rangle_{V,t}$ denotes a volume average over the whole domain combined with a temporal average in the statistically stationary state. Integrating and then averaging (10d), one can notice that:

$$\langle \partial_Z \bar{\theta} |_{Z=0} \rangle_t = \langle \partial_Z \bar{\theta} |_{Z=1} \rangle_t = -Nu, \quad (15)$$

where $\langle \cdot \rangle_t$ is a temporal average in the statistically stationary regime.

The usual Reynolds number $Re \stackrel{def}{=} U_{\text{RMS}} H / \nu$, where U_{RMS} is the dimensional root-mean-squared (RMS) velocity of the flow, can be computed using the ARM variables, according to:

$$Re = E^{-1/3} \sqrt{\langle |u'|^2 \rangle_{V,t} + \langle |v'|^2 \rangle_{V,t} + \langle |w'|^2 \rangle_{V,t}}, \quad (16)$$

where the factor $E^{-1/3}$ reflects the ARM nondimensionalization, in which the velocity unit is $E^{1/3} \times \nu / H$. A direct implication of those two definitions is the following crucial property:

- In the asymptotic rapidly rotating regime described by the ARM:

$$Nu \sim \mathcal{F}(\widetilde{Ra}, Pr) \quad \text{and} \quad Re \sim \mathcal{G}(\widetilde{Ra}, Pr) \times E^{-1/3}, \quad (17)$$

where $\mathcal{F}$ and $\mathcal{G}$ are unknown functions.

The ultimate regime of convection is defined as the fully turbulent scaling regime in which the large-scale properties of the flow no longer depend on the small molecular diffusivities, in accordance with the inertial dissipation laws of turbulence. Consequently, if we request that the scaling predictions (17) take the form of power laws independent of ν and κ , the scaling relations of the ultimate regime of rapidly rotating convection for the Nusselt and Reynolds numbers can be written as:

$$Nu \sim Ra^{3/2} E^2 Pr^{-1/2} = \widetilde{Ra}^{3/2} Pr^{-1/2}, \quad (18)$$

$$Re \sim Ra E Pr^{-1} = \widetilde{Ra} E^{-1/3} Pr^{-1}. \quad (19)$$

The derivation and comprehensive exposition of these scaling laws can be found in extant literature [6, 10, 27]. In the following, to simplify collapse of the data points, we will focus on the reduced Reynolds number $\widetilde{Re} = Re E^{1/3}$, which scaling prediction is:

$$\widetilde{Re} \stackrel{def}{=} Re E^{1/3} \sim \widetilde{Ra} Pr^{-1}. \quad (20)$$

2.4 Single-mode solutions

The objective of this study is to identify steady invariant solutions of the ARM, which can provide insight into the scaling properties of the ultimate regime of rapidly rotating Rayleigh–Bénard convection. As a first step in this direction, we must note that the ARM admits a particularly special class of steady solutions, referred to as single-mode solutions. These solutions, which have been analyzed in several previous studies [26, 28, 29], are obtained by applying a separation of variables of the form:

$$\text{For } \chi \in (\psi, \phi, \theta'), \quad \chi(x, y, Z, t) = h(x, y) \hat{\chi}(Z) \quad \text{with} \quad \begin{cases} \nabla_{\perp}^2 h &= -k^2 h \\ h^2 &= 1 \end{cases}, \quad (21)$$

where k is an arbitrary real number, that is analog to a wavenumber. Substituting this ansatz into the ARM equations (10) shows that an exact steady solution of this form exists and that it is governed solely by the following nonlinear ordinary differential equation:

$$d_{ZZ}\hat{\phi} + \left(\frac{\widetilde{Ra}Nu k^2}{1 + Pr^2 k^2 \hat{\phi}^2} - k^6 \right) \hat{\phi} = 0, \quad \text{with } \hat{\phi}(0) = \hat{\phi}(1) = 0, \quad (22)$$

where the Nusselt number is given by:

$$Nu = \left(\int_0^1 \frac{1}{1 + k^2 Pr^2 \hat{\phi}^2} dZ \right)^{-1}. \quad (23)$$

The other variables are functions of $\hat{\phi}$, and can be computed using the following relations:

$$\hat{\psi} = \frac{1}{k^2} d_Z \hat{\phi}, \quad (24a)$$

$$\hat{\theta}' = \frac{Pr}{\widetilde{Ra} k^2} \left(k^6 \hat{\phi} - d_{ZZ} \hat{\phi} \right) = \frac{Pr Nu \hat{\phi}}{1 + Pr^2 k^2 \hat{\phi}^2}, \quad (24b)$$

$$\partial_Z \bar{\theta} = \frac{1}{\widetilde{Ra} k^2 \hat{\phi}} \left(d_{ZZ} \hat{\phi} - k^6 \hat{\phi} \right) = -\frac{Nu}{1 + Pr^2 k^2 \hat{\phi}^2}. \quad (24c)$$

Additionally, introducing the change of variables $\hat{\phi}_* = Pr \hat{\phi}$ eliminates the explicit dependence on Pr in equation (22). Consequently, $\widetilde{Ra}$ and k are the only parameters relevant for single-mode solutions.

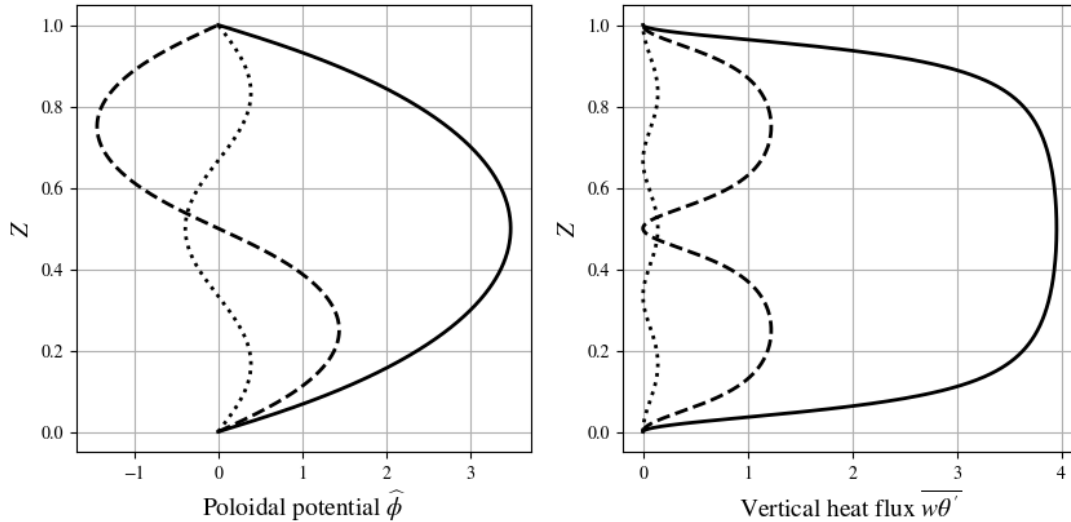


Figure 1: Poloidal potential and vertical heat flux profiles for three single-mode solutions with $k = 1$ and $\widetilde{Ra}_P = 100$. The solid line denotes the fundamental solution, while the dotted and dashed lines represent harmonic solutions. Only the fundamental solution exhibits a physically realistic vertical heat flux profile.

To solve this equation, we fix the wavenumber k and the parameter $\widetilde{Ra}_P = \widetilde{Ra}Nu$, treating $\widetilde{Ra}$ as an output rather than a control parameter. The Rayleigh number is then recovered by dividing $\widetilde{Ra}_P$ by the Nusselt number (23), which is also determined a posteriori. A shooting method is employed to solve equation (22) numerically for the parameter pair $(k, \widetilde{Ra}_P)$ while enforcing Dirichlet boundary conditions at $Z = 0$ and $Z = 1$. Because of the strong nonlinearities, equation (22) may admit multiple solutions for the same parameter set $(k, \widetilde{Ra}_P)$. First, the conductive state $\hat{\phi}(z) = 0$ is always solution. However,

more complex nonlinear solutions can also be found. In particular, the finite vertical extension induces the existence of one fundamental solution and several harmonic solutions, analogous to the modes of a vibrating string. The vertical velocity of the fundamental solution exhibits a single maximum at $z = 1/2$ and vanishes only at the boundaries, whereas the velocity profiles of the harmonics contain multiple zeros. Since this latter behavior is physically unrealistic, only the fundamental solution will be considered. A selection of possible solutions for the parameter set $k = 1$ and $\widetilde{Ra}_P = 100$ is plotted on figure 1. In the following, the term ‘single-mode solution’ will refer to the fundamental solution of equation (22).

By varying the wavenumber k , one generates a constellation of fixed points in the phase space, although not all of them are necessarily relevant for describing the actual trajectory of the chaotic solution. Therefore, we decide to track a continuous branch of steady solutions starting from the onset of convection. Our objective is to characterize the scaling properties of the Nusselt and Reynolds numbers along this branch. Indeed, since this branch originates at a supercritical bifurcation [30–32], it is relevant for describing the transport properties of the actual solution at a finite distance from the threshold. However, even though these properties may not remain valid farther from the onset, they can leave a signature on the statistical features of the chaotic solution—particularly if this continuous branch of steady solutions acts as a fixed point attracting the chaotic trajectory. In this study, single-mode solutions serve both as a reference and as a tool for characterizing the continuous branch of steady solutions that we track.

3 A continuous branch of steady solutions

3.1 Numerical procedure

Let $\underline{R}(t) = \{\psi(t), \phi(t), \theta'(t), \bar{\theta}(t)\}$ be a position vector in the phase space. A steady solutions may be defined as a point in the phase space such that:

$$\underline{R} - \varphi_\tau(\underline{R}) = 0 \quad \text{for every time } \tau > 0, \quad (25)$$

with $\varphi_\tau(\underline{R}(t)) = \underline{R}(t + \tau)$ is the flow function.

The Newton-Krylov method such as described by Motoki et al. [33] aims to find a numerical approximation to the solution of the problem (25). To do so, fields are discretized on a Chebyshev-Fourier-Fourier space grid, while the flow function is numerically estimated with DNS, using the pseudo-spectral time-stepper Coral [34]. To ensure convergence, the Newton-Krylov method requires to start from an initial guess that is relatively close to the exact solution.

Just above the onset of convection, the system eventually reaches a steady state after some transient phase. Starting from this steady state, we slightly increase $\widetilde{Ra}$ and apply the Newton-Krylov method, using a DNS snapshot of the steady state as the initial guess. This procedure yields a new steady solution at the slightly larger $\widetilde{Ra}$. Repeating this process—incrementing $\widetilde{Ra}$ and reusing the previous steady solution as the initial guess—allows us to track a continuous branch of steady solutions.

3.2 Field descriptions

Application of the tracking method for $Pr = 1$ and constant aspect ratio $L/H = 48.154\varepsilon$, yields a continuous branch of steady solutions, the properties of which are examined in this section.

Horizontal and vertical slices of temperature fluctuations for increasing $\widetilde{Ra}$ are shown in figure 2. The flow is organized into vertical columns of nearly uniform size, distributed horizontally in a complex pattern. Notably, no small-scale repeating horizontal structures can be identified. Increasing $\widetilde{Ra}$ induces a vertical elongation of the columns, while no significant variation is observed in the horizontal direction. Figure 3 shows horizontal and vertical slices of velocity fluctuations for $\widetilde{Ra} = 10.3$. The vertical velocity w' exhibits the same pattern as the temperature fluctuations, whereas the horizontal velocity components display slight differences. In particular, the vertical slices reveal that the horizontal velocity amplitudes are maximal at the boundaries, while the vertical velocity vanishes there.

3.3 Solution identification

The continuous branch of steady solutions we tracked exhibits a striking horizontal pattern that remains unchanged with increasing $\widetilde{Ra}$. Moreover, the columnar structure suggests a spatial separation between the horizontal and vertical directions. This raises the question of the relationship between this branch and single-mode solutions. To address this, we first extract a typical horizontal wave number K for every

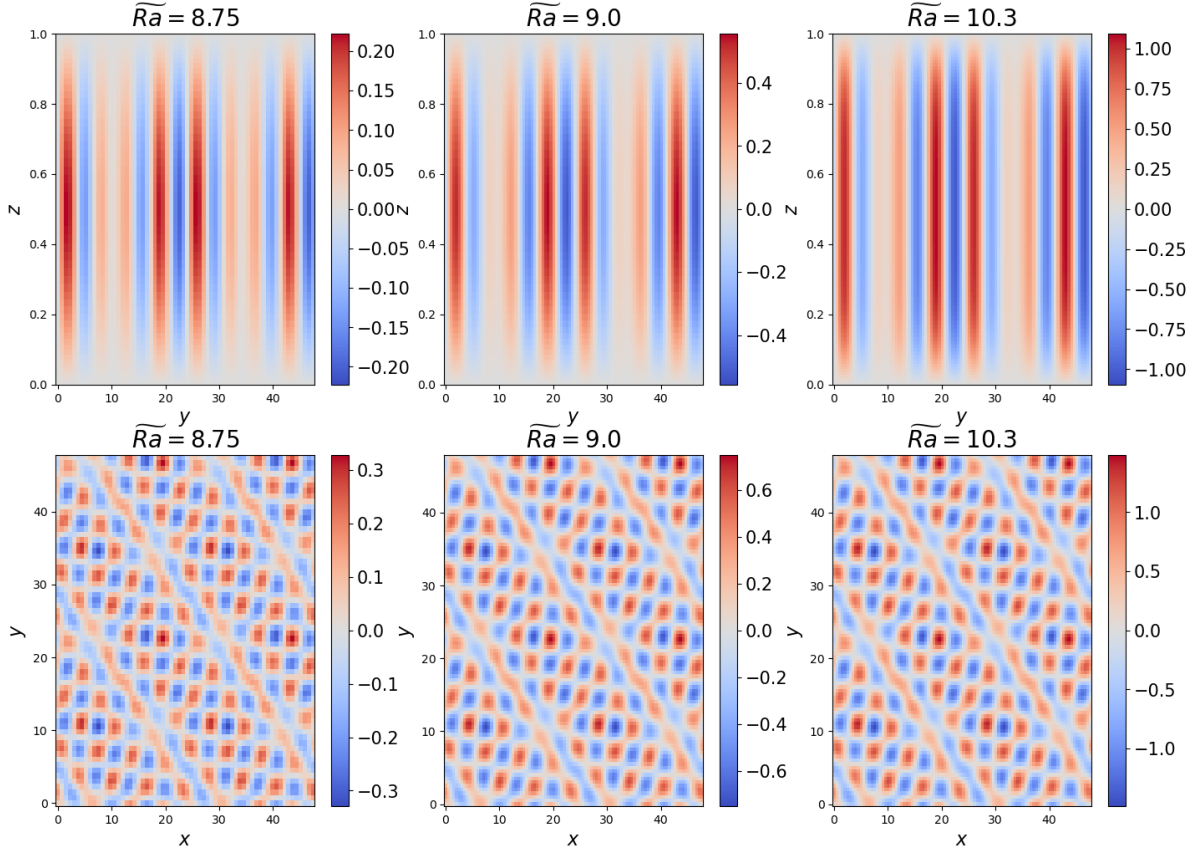


Figure 2: Vertical and mid-depth horizontal slices of temperature fluctuations θ' for three solutions of the continuous branch of steady solutions we tracked.

numerical steady solution, by computing:

$$K = \left\langle \frac{\langle (\nabla_{\perp}^2 \phi)^2 \rangle_{x,y}}{\langle \phi^2 \rangle_{x,y}} \right\rangle_Z^{1/4}, \quad (26)$$

where $\langle \cdot \rangle_{x,y}$ is a horizontal average and $\langle \cdot \rangle_Z$ is a vertical average. For all our steady solutions we find $K = 1.304810 \pm 10^{-6}$. However, it appears that $k_c \approx 1.3048$ is the marginal wavenumber at the onset of convection [35]. Therefore, we can state that along our continuous branch of steady solutions, the typical horizontal wavenumber remains constant and equal to the critical value k_c .

A steady solution is a single-mode solution if there exists a wavenumber k such that, for every field $\chi \in \{\psi, \phi, \theta'\}$, the relation $\nabla_{\perp}^2 \chi + k^2 \chi = 0$ holds. Figure 4 shows a horizontal slice at mid-depth of the quantity $\nabla_{\perp}^2 \theta' + K^2 \theta'$ for $\widetilde{Ra} = 9$. One can notice that this quantity is negligible compared with $\nabla_{\perp}^2 \theta'$, whose horizontal slice is plotted alongside for comparison. This observation suggests that our steady solution corresponds to a single-mode solution with $k = K$. More generally, one can verify whether a steady solution is single-mode with $k = K$, by computing the error:

$$\eta = \frac{1}{3} \sum_{\chi \in \mathcal{V}} \left[\frac{\langle (\nabla_{\perp}^2 \chi + K^2 \chi)^2 \rangle_V}{\langle (\nabla_{\perp}^2 \chi)^2 \rangle_V} \right]^{1/2} \quad \text{with } \mathcal{V} = \{\psi, \phi, \theta'\}. \quad (27)$$

This error is evaluated for every steady solution along the tracked branch. In all cases, the error η remains below 6×10^{-3} . This demonstrates that the continuous branch of steady solutions we tracked is, in fact, composed of single-mode solutions with a constant wavenumber $k = k_c$. Therefore, we can now exploit theoretical results on single-mode solutions to characterize this branch.

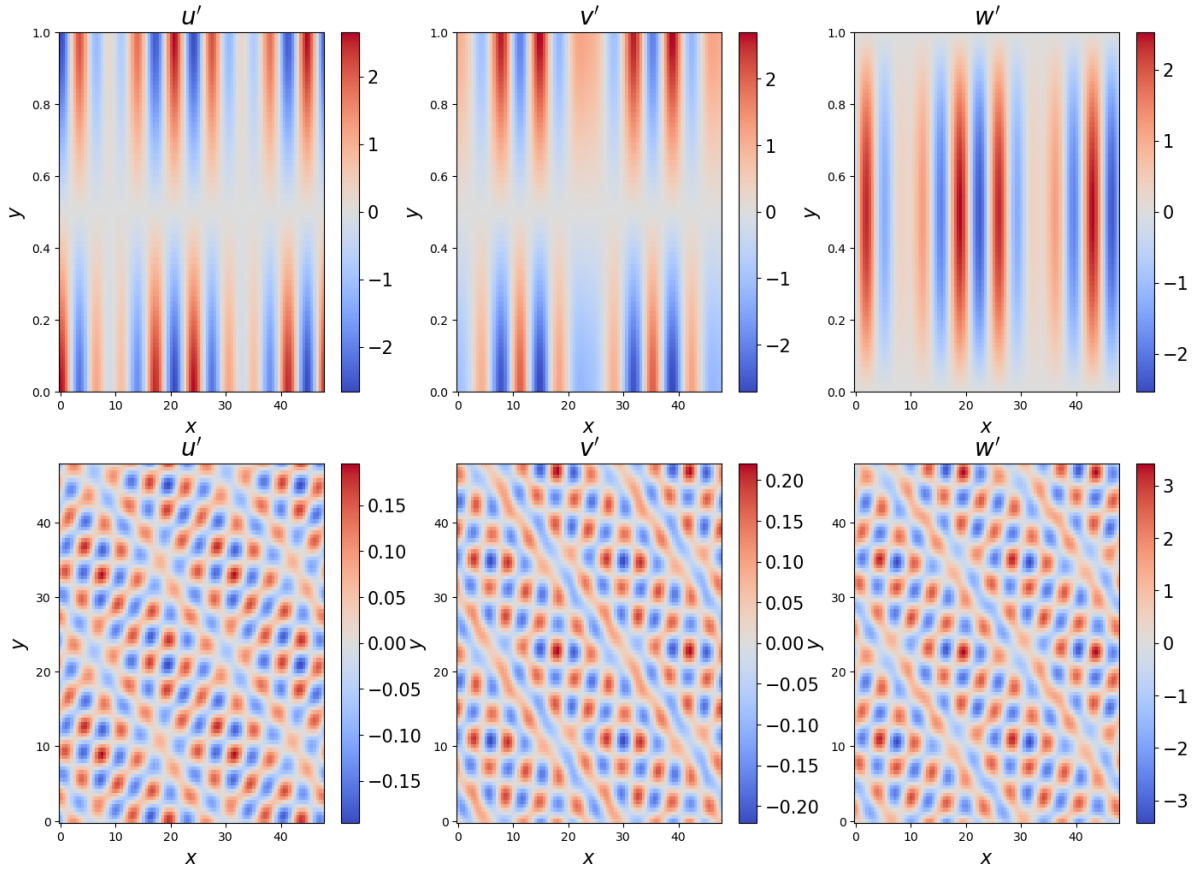


Figure 3: Vertical and mid-depth horizontal slices of the three components of velocity fluctuations for $\widetilde{Ra} = 10.3$.

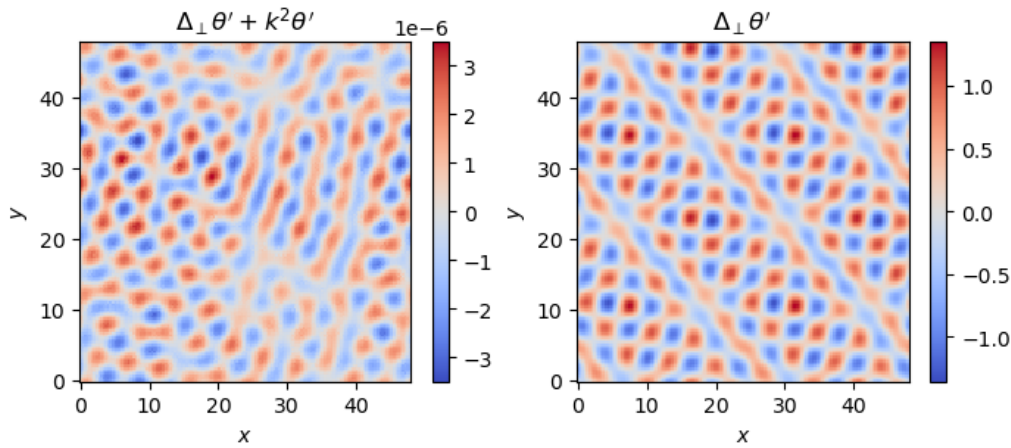


Figure 4: Mid-depth horizontal slice of the quantity $\nabla_{\perp}^2 \theta' + K^2 \theta'$ for $\widetilde{Ra} = 9$. A mid-depth horizontal slice of $\nabla_{\perp}^2 \theta'$ is plotted on the right side for comparison.

4 Scaling behavior

4.1 Theoretical predictions

In this section, we use the fact that our continuous branch is composed of single-mode solutions to derive theoretical scaling properties that hold along it. We begin by writing the energy budget of the ARM. Multiplying the equations of system (10) by the appropriate variables, and then taking a spatial average yields:

$$\frac{d}{dt} \left(\frac{\langle (\nabla_{\perp} \psi)^2 \rangle_V}{2} + \frac{\langle (\nabla_{\perp} \phi)^2 \rangle_V}{2} \right) = \underbrace{\frac{\widetilde{Ra} \langle w' \theta' \rangle_V}{Pr}}_{\text{Energy injected by buoyancy}} - \underbrace{\left(\langle \zeta^2 \rangle_V + \langle (\nabla_{\perp} w)^2 \rangle_V \right)}_{\text{Viscous dissipation}}. \quad (28)$$

Since single-mode solutions are steady exact solutions to the ARM, the energy budget becomes:

$$\frac{\widetilde{Ra} (Nu - 1)}{Pr^2} = \langle \zeta^2 \rangle_{V,t} + \langle (\nabla_{\perp} w)^2 \rangle_{V,t}. \quad (29)$$

However, leveraging horizontal periodicity: $\langle \zeta^2 \rangle_{V,t} = k^2 \langle (\nabla_{\perp} \psi)^2 \rangle_{V,t} = k^2 \langle u'^2 + v'^2 \rangle_{V,t}$. Our branch of steady solutions then satisfies:

$$\frac{\widetilde{Ra} (Nu - 1)}{Pr^2} = k_c^2 \widetilde{Re}^2. \quad (30)$$

This equation establishes the mutual dependence between the Nusselt and Reynolds numbers along the continuous branch of steady solutions we tracked. In particular, if the Nusselt number follows a diffusivity-free scaling law, as suggested by the definition of the ultimate regime, $Nu \sim \widetilde{Ra}^{3/2} Pr^{-1/2}$, then equation (30) yields:

$$\widetilde{Re} \sim \left(\widetilde{Ra} / Pr \right)^{5/4}. \quad (31)$$

This scaling law differs from the diffusivity-free scaling proposed for the Reynolds number (20). Thus, this exact relation highlights the incompatibility between diffusivity-free scaling laws for the Nusselt and Reynolds numbers along the tracked branch. A natural question then arises: does the incompatibility reported in the literature for chaotic solutions follow the same incompatibility relation that we have identified for the continuous branch of steady solutions? This question is appropriate taking into consideration the branch we track originates at a supercritical bifurcation [30–32], where the incompatibility relation is then rigorously valid. It is therefore reasonable to ask whether this relation holds as we increase $\widetilde{Ra}$ to higher values.

4.2 Results

In this section, we perform DNS of both the ARM and the full system of equations for $Pr = 1$ using the pseudo-spectral solver Coral [34]. Our goal is to compare this diagnostic valid along our continuous branch of steady solutions with the actual properties of the chaotic solution. A distinctive feature of stress-free RRBC is the presence of an inverse kinetic energy cascade when convection becomes sufficiently vigorous [36–40]. This process leads to the formation of large-scale horizontal vortices that grow until they reach the size of the computational domain. In such cases, the Reynolds number based on the total RMS velocity no longer provides a reliable measure of the convective kinetic energy, as it becomes dominated by the two-dimensional spectral condensation. Consequently, and in agreement with several previous studies [15, 37, 39, 41], we instead consider the vertical Reynolds number Re_z , defined from the vertical RMS velocity, which offers a more appropriate measure of convective intensity. We also introduce its reduced expression, $\widetilde{Re}_z = Re_z E^{1/3}$. Because of the inverse cascade, the horizontal kinetic energy is greatly dependent on the domain aspect ratio. However, it has a weak influence on both Nu and Re_z [37, 39].

Figures 5 and 6 show the Nusselt number and the reduced vertical Reynolds number obtained from our DNS as functions of the reduced Rayleigh number. A first observation is that the data points for RRBC collapse onto a single master curve in these representations, independently of the Ekman number, in agreement with the ARM. At large Rayleigh numbers, the Nusselt number follows a diffusivity-free scaling law for both the ARM and stress-free RRBC, as reported in the literature [15, 42]. In agreement with the observations of Kannan et al. [15], the Reynolds number deviates slightly from the diffusivity-free scaling prediction (20). In contrast, the diagnosed relation (31), shown as a dashed line in figure 6,

exhibits excellent agreement with the DNS data. The difference between the exponents 1 and $5/4$ may appear small, but this deviation becomes clearly visible on the compensated figures, especially for the ARM. In conclusion, these results provide evidence that the incompatibility relation characterizing the continuous branch of steady solutions also holds, to a good approximation, for the turbulent solution.

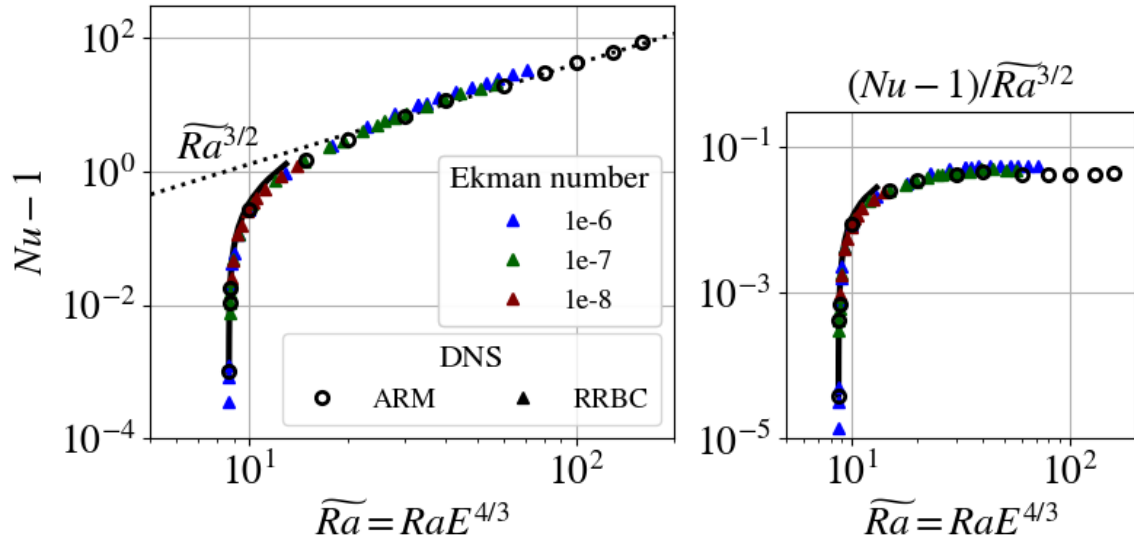


Figure 5: Dimensionless convective heat flux $(Nu - 1)$ as a function of the reduced Rayleigh number $\widetilde{Ra}$ from DNS of the ARM and DNS of RRBC with stress-free boundary conditions. The dotted line indicates the diffusivity-free scaling for the Nusselt number (18), while the black solid line represents the Nusselt number along the tracked branch of steady solutions. The figure on the right shows the Nusselt number compensated by the diffusivity-free prediction.

5 Concluding remarks

Using the Newton–Krylov numerical method, we have tracked a continuous branch of steady invariant solutions of the ARM for fixed Prandtl number $Pr = 1$. An analysis of their spatial structure reveals that all of these solutions correspond to single-mode solutions with a constant wavenumber $k = k_c$. This finding allows us to establish that, along this branch, the Reynolds and Nusselt numbers are linked through the exact relation (30). Consequently, if the Nusselt number follows a diffusivity-free scaling law on this branch, the Reynolds number must then satisfy:

$$\widetilde{Re} \sim \left(\widetilde{Ra}/Pr \right)^{5/4}. \quad (32)$$

This scaling relation, theoretically valid for our branch of steady solutions, actually provides an accurate scaling prediction for the fully chaotic state, as confirmed by DNS of both the ARM and stress-free RRBC. This result raises several questions about the mechanisms that could explain such an agreement. In particular, the dominant convective length scale appears to play a central role in shaping the transport-scaling properties of rotating convection. For example, Kannan et al. [15] propose theoretical arguments supporting a very similar exponent, namely $11/8$, based on an analysis of the characteristic length scales associated with the large-scale circulation that develops at high Rayleigh number. In their framework, the typical convective length scale increases with thermal forcing as $\widetilde{Ra}^{1/8}$. In contrast, along our branch of steady solutions, the characteristic length scale entering the exact relation (30) is simply the wavelength of the horizontal pattern, which remains constant as $\widetilde{Ra}$ increases. In summary, rotating convection involves multiple interacting length scales, and determining which of them govern the transport properties remains an important open question for future work.

On a related note, the same relationship (30) that is based on the energy budget of single-mode solutions, actually contains additional information about length scales. For a family of single-mode

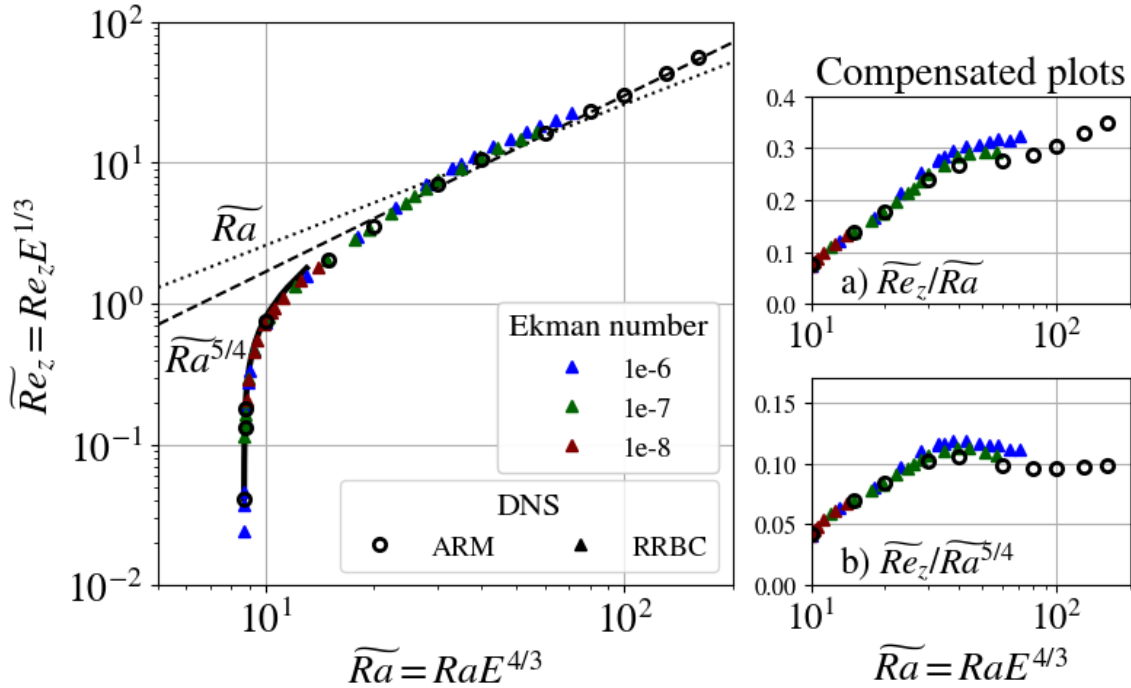


Figure 6: Reduced vertical Reynolds number $\widetilde{Re}_z$ as a function of the reduced Rayleigh number $\widetilde{Ra}$ from DNS of the ARM and DNS of RRBC with stress-free boundary conditions. The black solid line represents $\widetilde{Re}_z$ along the tracked branch of steady solutions. The dotted line denotes the diffusivity-free scaling for the Reynolds number (20), while the dashed line shows the scaling prediction derived from the properties of the continuous branch of steady solutions (31). The two right-hand figures show $\widetilde{Re}_z$ compensated by the diffusivity-free prediction (top) and by the scaling prediction derived from the continuous branch of steady solutions (bottom).

solutions with $k \sim \widetilde{Ra}^\gamma Pr^\delta$, the Reynolds number is related to the Nusselt number through the scaling relation:

$$\widetilde{Re} \sim Nu^{1/2} \widetilde{Ra}^{1/2-\gamma} Pr^{-1-\delta}. \quad (33)$$

This expression allows simultaneous diffusivity-free scaling laws for the Nusselt and Reynolds numbers only if $\gamma = 1/4$ and $\delta = -1/4$. This result is consistent with the study by Grooms [29] on theoretical bounds for heat transport in single-mode solutions, in which he showed that single-mode solutions with $k \sim \widetilde{Ra}^{1/4}$ maximize the lower and upper bounds on the heat flux, allowing $Nu \sim \widetilde{Ra}^{3/2}$. If single-mode solutions remain relevant for describing the scaling behavior of chaotic solutions at higher Rayleigh numbers, then this would imply that the typical horizontal scale must decrease with $\widetilde{Ra}$ in order for the system to reach a diffusivity-free regime for both the Nusselt and Reynolds numbers. However, based on the behavior of the characteristic length scales reported in the literature [10, 11, 15] and the emergence of large-scale circulation at high $\widetilde{Ra}$ [36, 37], such a decrease appears unlikely even at significantly higher Rayleigh and smaller Ekman numbers. This raises fundamental questions about whether a fully diffusivity-free turbulent regime can actually exist. Finally, recent work by van Kan et al. [41] offers promising prospects for addressing this question. Their newly rescaled RRBC model [43] makes it possible to explore Ekman numbers that are orders of magnitude smaller than those currently attainable in DNS of the full RRBC system. Accessing such extreme parameter regimes should help clarify how the Reynolds number behaves at very large $\widetilde{Ra}$ and, more broadly, whether the system truly approaches a diffusivity-free asymptotic state.

Acknowledgments

This work was supported by JSPS KAKENHI Grant Numbers JP24KF0239, JP25K01157, JP25K07567, and by the European Research Council under the Caust grant ERC-AdG-101018287. G. Hadjerci acknowledges financial support as a Japan Society for the Promotion of Science International Research Fellow. This research used the supercomputer system of Academic Center for Computing and Media Studies, Kyoto University. The authors would like to thank J. Jimenez and A. Guseva for helpful discussions and suggestions that have improved the manuscript.

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